Mobile Application Development

(DI04016051)

LABORATORY MANUAL

Diploma Sem- 4

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Academic Year : 2025-2026

Index

Software Requirements

Sr No	Software Requirement	Hardware Requirement
1	Android Studio	64-bit OS
		RAM-16 GB
		Processor (Intel i5/i7, AMD Ryzen, etc.)
		Hard Disk- 512 GB

Date:

PRACTICAL 1

AIM: Write Android program to display "Welcome to android".

```
<!-- res/layout/activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to android"
        android:textSize="24sp"
        android:textStyle="bold"/>
  </LinearLayout>
```

Output:

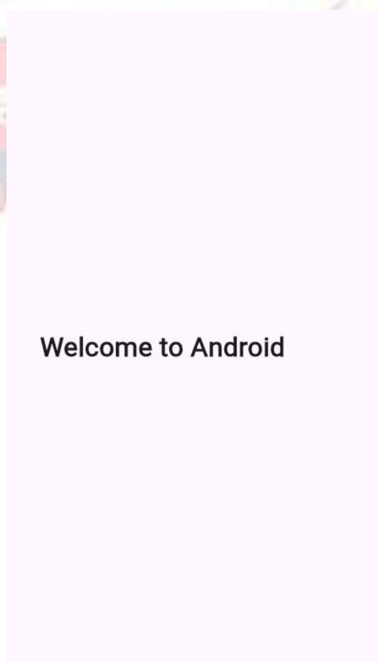

Fig 1.1 Output

Logical Application

1. **What is the default layout file used in an Android activity?**

2. **Which XML attribute sets the text displayed in a TextView?**

Date:

PRACTICAL 2

AIM: Write android program to demonstrate usage of string.

Program:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:background="#FFFFFF">

    <TextView
        android:id="@+id/textViewResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="String Example"
        android:textSize="22sp"
        android:textColor="#000000" />
</LinearLayout>
```

Main Activity.java

```
    package com.example.stringdemo;

    import androidx.appcompat.app.AppCompatActivity;
    import android.os.Bundle;
    import android.widget.TextView;

    public class MainActivity extends AppCompatActivity {

        @Override
        protected void onCreate(Bundle savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_main);

            TextView textView = findViewById(R.id.textViewResult);
```

```
        // String usage demonstration
        String firstName = "Android";
        String lastName = "Programming";

        // Concatenation
        String fullName = firstName + " " + lastName;

        // Display result
        textView.setText("Concatenated String: " + fullName);
    }
}
```

Output:

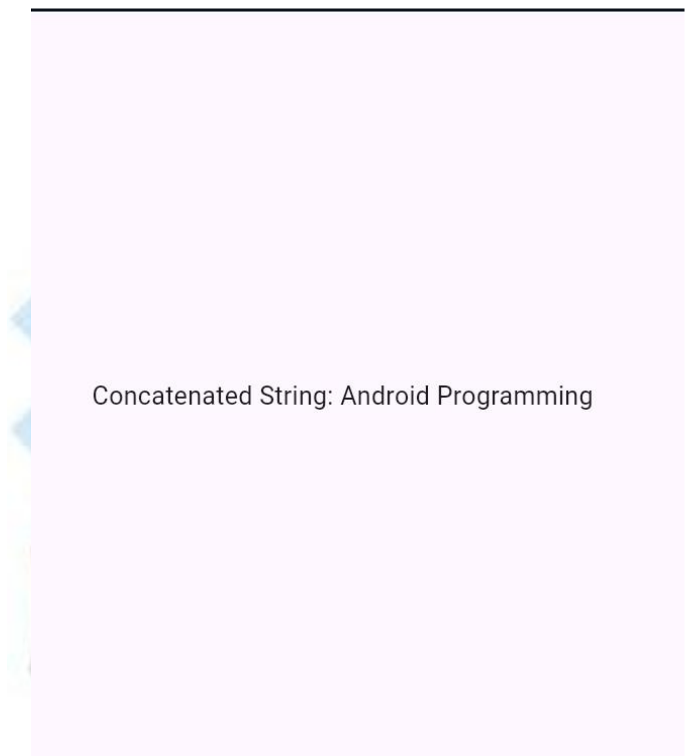

Fig 2.1 Output

Logical Application

1. **What class in Android is used to store string resources?**

2. **How do you access a string resource in Java using its ID?**

Date:

PRACTICAL 3

AIM: Write android program to demonstrate activity life cycle.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#FFFFFF"
    android:padding="20dp">

    <TextView
        android:id="@+id/textViewLifeCycle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Life Cycle Demo"
        android:textSize="20sp"
        android:textColor="#000000"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.lifecycleapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    private static final String TAG = "ActivityLifeCycle";
    TextView textView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        textView = findViewById(R.id.textViewLifeCycle);
        textView.setText("onCreate() called");
        Log.d(TAG, "onCreate() called");
    }
```

```
    @Override
    protected void onStart() {
        super.onStart();
        Log.d(TAG, "onStart() called");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d(TAG, "onResume() called");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause() called");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop() called");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart() called");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy() called");
    }
}
```

Output:

Fig 3.1 Output

Logical Application

1. **Which method is used to set the background color of a layout programmatically?**

2. **What XML attribute sets the background color of a View?**

Date:

PRACTICAL 4

AIM: Write android program to change the background of activity.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:id="@+id/mainLayout">

    <Button
        android:id="@+id/btnChangeColor"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Background" />
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.backgrounddemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.graphics.Color;
import android.view.View;
import android.widget.Button;
import android.widget.LinearLayout;
public class MainActivity extends AppCompatActivity {
    LinearLayout mainLayout;
    Button btnChange;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mainLayout = findViewById(R.id.mainLayout);
        btnChange = findViewById(R.id.btnChangeColor);
        btnChange.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Change background color on button click
                mainLayout.setBackgroundColor(Color.CYAN);
            }
```

```
        });
    }
}
```

Output :

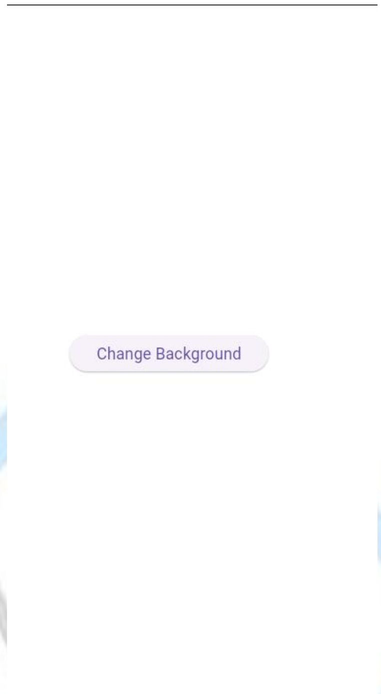

Fig 4.1 Output

Logical Application

1. **Name the first method called when an Android activity is created.**

____

____

____

2. **Which lifecycle method is called when an activity becomes invisible to the user?**

____

____

____

Date:

PRACTICAL 5

AIM: Write android program to perform all operation in calculator.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center"
    android:id="@+id/mainLayout">

    <EditText
        android:id="@+id/etNum1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
        android:inputType="numberDecimal"
        android:layout_marginBottom="10dp"/>

    <EditText
        android:id="@+id/etNum2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center"
        android:layout_marginBottom="20dp">

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="+" />
```

```
        <Button
            android:id="@+id/btnSub"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="-"
            android:layout_marginLeft="10dp"/>

        <Button
            android:id="@+id/btnMul"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="×"
            android:layout_marginLeft="10dp"/>

        <Button
            android:id="@+id/btnDiv"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="÷"
            android:layout_marginLeft="10dp"/>
    </LinearLayout>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will be shown here"
        android:textSize="20sp"
        android:textColor="#000000"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.calculatordemo;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
```

```
EditText etNum1, etNum2;
Button btnAdd, btnSub, btnMul, btnDiv;
TextView tvResult;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etNum1 = findViewById(R.id.etNum1);
    etNum2 = findViewById(R.id.etNum2);
    btnAdd = findViewById(R.id.btnAdd);
    btnSub = findViewById(R.id.btnSub);
    btnMul = findViewById(R.id.btnMul);
    btnDiv = findViewById(R.id.btnDiv);
    tvResult = findViewById(R.id.tvResult);

    btnAdd.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            calculate('+');
        }
    });

    btnSub.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            calculate('-');
        }
    });

    btnMul.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            calculate('*');
        }
    });

    btnDiv.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            calculate('/');
        }
    });
```

```
    }

    private void calculate(char operator) {
        String num1Str = etNum1.getText().toString();
        String num2Str = etNum2.getText().toString();

        if (num1Str.isEmpty() || num2Str.isEmpty()) {
            Toast.makeText(this, "Please enter both numbers", Toast.LENGTH_SHORT).show();
            return;
        }

        double num1 = Double.parseDouble(num1Str);
        double num2 = Double.parseDouble(num2Str);
        double result = 0;

        switch (operator) {
            case '+':
                result = num1 + num2;
                break;
            case '-':
                result = num1 - num2;
                break;
            case '*':
                result = num1 * num2;
                break;
            case '/':
                if (num2 != 0) {
                    result = num1 / num2;
                } else {
                    Toast.makeText(this, "Cannot divide by zero", Toast.LENGTH_SHORT).show();
                    return;
                }
                break;
        }

        tvResult.setText("Result: " + result);
    }
}
```

OUTPUT :

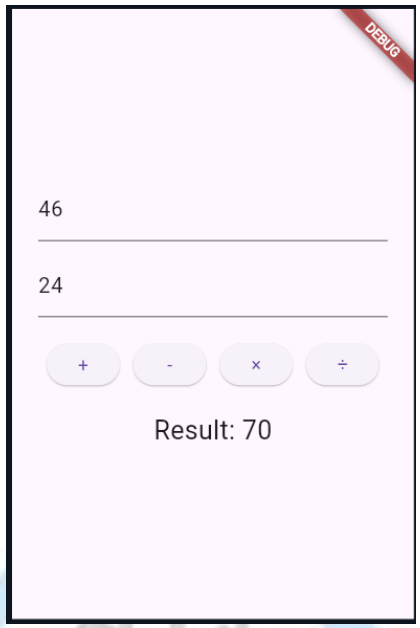

Fig 5.1 Output

Logical Application

1. **Which listener is used to handle button clicks in a calculator app?**

2. **What data type is best suited for storing decimal calculator results in Java?**

Date:

PRACTICAL 6

AIM: Write android program to create multiple activities within an application.

Program:

XML Layout for MainActivity (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnNext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity" />
</LinearLayout>
```

XML Layout for SecondActivity (activity_second.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="20sp"
        android:textColor="#000000"/>
</LinearLayout>
```

Java Code for MainActivity.java

```
package com.example.multipleactivities;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    Button btnNext;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Intent to open SecondActivity
                Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

Java Code for SecondActivity.java

```
package com.example.multipleactivities;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

AndroidManifest.xml (Add Second Activity)

```
<application
    android:allowBackup="true"
    android:icon="@mipmap/ic_launcher"
```

```
    android:label="@string/app_name"
    android:supportsRtl="true"
    android:theme="@style/Theme.AppCompat.Light.NoActionBar">

    <activity android:name=".SecondActivity"></activity>
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Output :

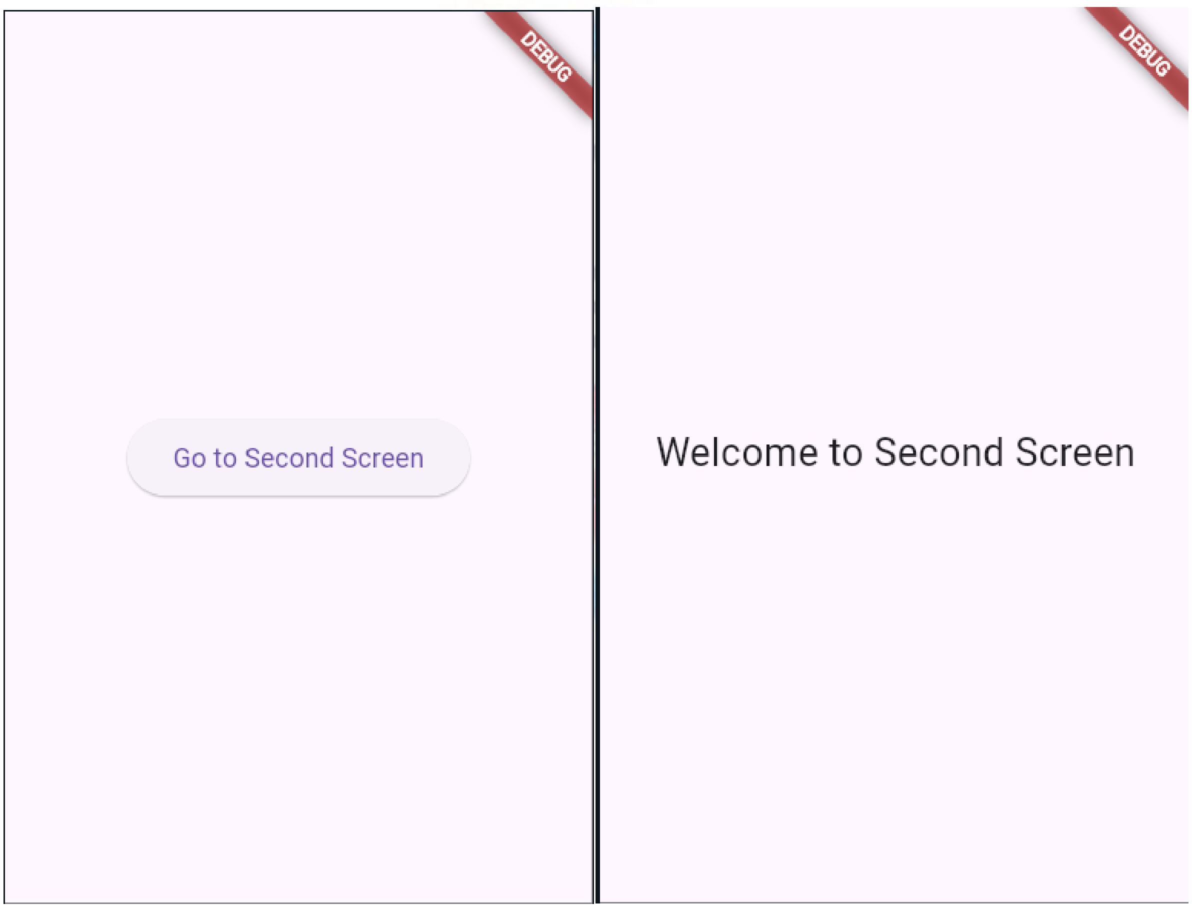

Fig 6.1 Output

Logical Application

1. **Which class is used to navigate from one activity to another in Android?**

2. **What method is called to start a new activity using an Intent?**

Date:

PRACTICAL 7

AIM: Write android program to demonstrate action button by implementing ONCLICKLISTENER.

<u>Program:</u>

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnClickMe"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.onclickdemo;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity implements View.OnClickListener {

    Button btnClickMe;
    TextView tvResult;
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    btnClickMe = findViewById(R.id.btnClickMe);
    tvResult = findViewById(R.id.tvResult);

    // Set OnClickListener
    btnClickMe.setOnClickListener(this);
}

@Override
public void onClick(View v) {
    if (v.getId() == R.id.btnClickMe) {
        tvResult.setText("Button Clicked Successfully!");
    }
```

Output :

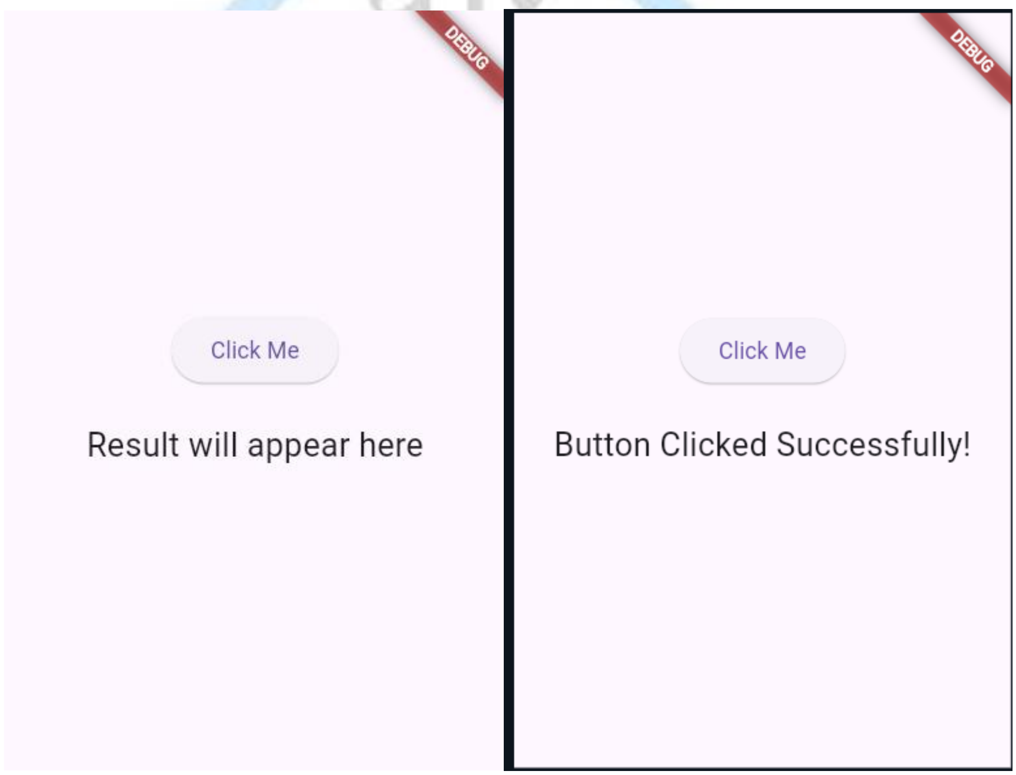

Fig 7.1 Output

Logical Application

1. **Which interface must be implemented to handle button click events?**

2. **What method do you override when implementing OnClickListener?**

Date:

PRACTICAL 8

AIM: Write android program to demonstrate sound button application.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <Button
        android:id="@+id/btnPlaySound"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Sound" />
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.soundbuttonapp;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button btnPlaySound;
    MediaPlayer mediaPlayer;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnPlaySound = findViewById(R.id.btnPlaySound);

        // Initialize MediaPlayer with sound file from res/raw
        mediaPlayer = MediaPlayer.create(this, R.raw.clicksound);
        btnPlaySound.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
            public void onClick(View v) {
                // Play sound when button clicked
                if (mediaPlayer != null) {
                    mediaPlayer.start();
                }
            }
        });
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        if (mediaPlayer != null) {
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}
```

Output :

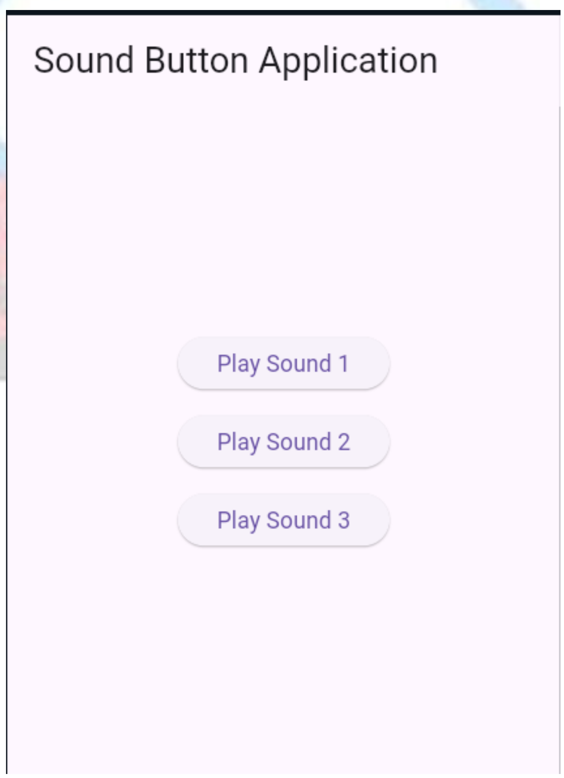

Fig 8.1 Output

Logical Application

1. **Which Android class is used to play short sound effects?**

2. **What method of MediaPlayer is called to begin audio playback?**

Date:

PRACTICAL 9

AIM: Write an Android application to convert into different currencies for example, Rupees to dollar.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etRupees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter amount in Rupees"
        android:inputType="numberDecimal"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert to Dollar" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.currencyconverter;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText etRupees;
    Button btnConvert;
    TextView tvResult;

    // Conversion rate (Example: 1 Rupee = 0.012 USD)
    final double DOLLAR_RATE = 0.012;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etRupees = findViewById(R.id.etRupees);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rupeesStr = etRupees.getText().toString();

                if (rupeesStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter amount in Rupees",
Toast.LENGTH_SHORT).show();
                    return;
                }

                double rupees = Double.parseDouble(rupeesStr);
                double dollars = rupees * DOLLAR_RATE;

                tvResult.setText("USD: $" + String.format("%.2f", dollars));
            }
        });
    }
}
```

Output :

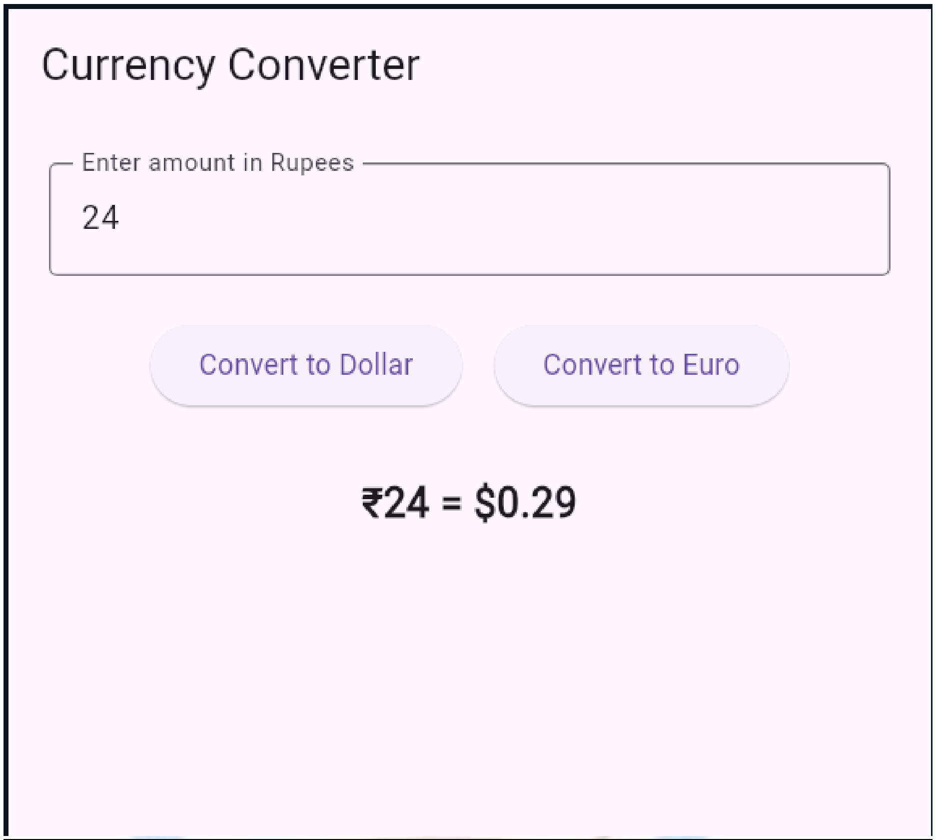

Fig 9.1 Output

Logical Application

1. **What layout widget is used to take numeric input from the user?**

2. **How do you convert a String from an EditText to a float in Java?**

Date:

PRACTICAL 10

AIM: Write an android application to count library overdue.

Program:
XML Layout (activity_main.xml)

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etDays"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter number of overdue days"
        android:inputType="number"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnCalculate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate Fine" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fine will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.libraryfine;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {

    EditText etDays;
    Button btnCalculate;
    TextView tvResult;

    // Fine rate per day
    final int FINE_RATE = 2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etDays = findViewById(R.id.etDays);
        btnCalculate = findViewById(R.id.btnCalculate);
        tvResult = findViewById(R.id.tvResult);

        btnCalculate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String daysStr = etDays.getText().toString();

                if (daysStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter overdue days",
Toast.LENGTH_SHORT).show();
                    return;
                }
                int days = Integer.parseInt(daysStr);
                int fine = days * FINE_RATE;

                tvResult.setText("Total Fine: ₹" + fine);
            }
        });
    }
}
```

Output :

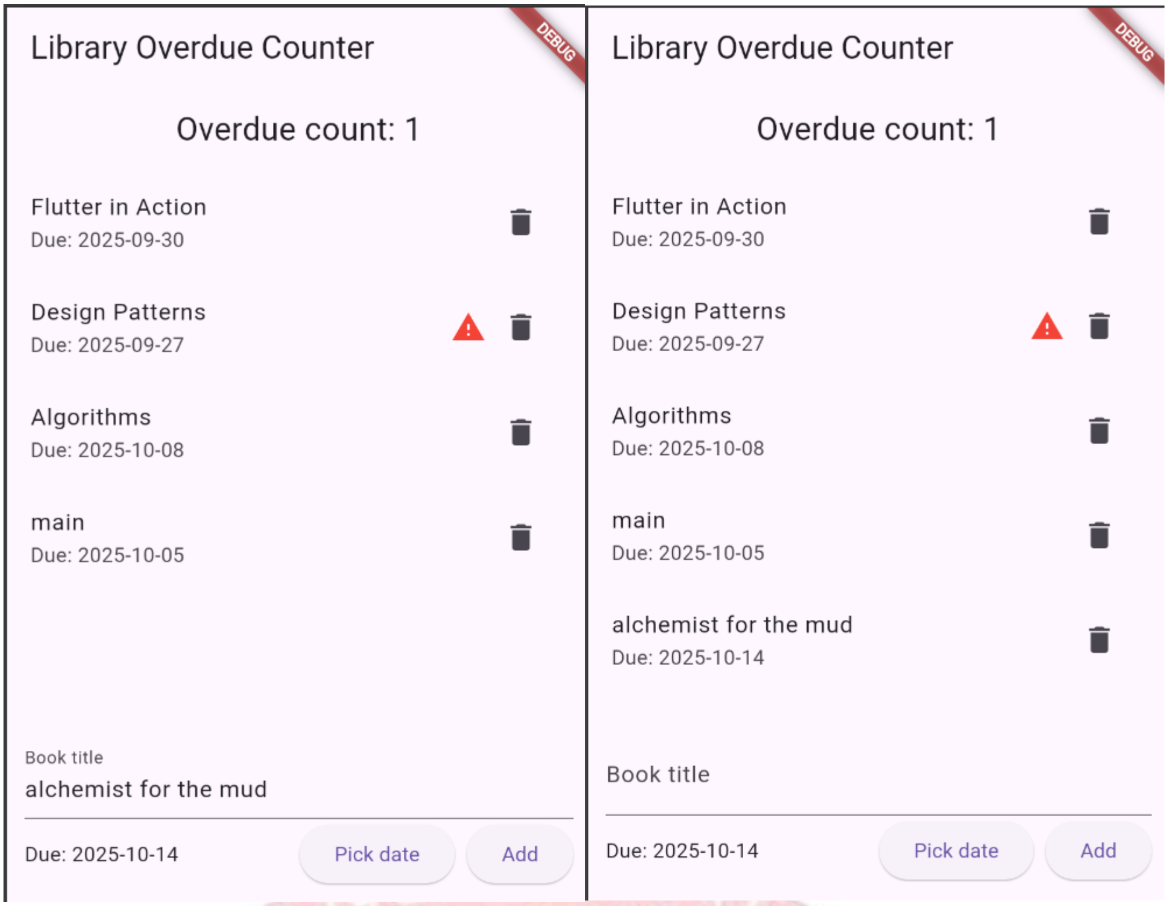

Fig 10.1 Output

Logical Application

1. **Which Java class is used to calculate the difference between two dates?**

____

____

____

2. **What widget is used to let the user pick a date in Android?**

____

____

____

Date:

PRACTICAL 11

AIM: Write an application to mark the daily route of travel in map.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<fragment xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
android:layout_height="match_parent"/>
```

Java Code (MainActivity.java)

```
package com.example.dailyroute;
import androidx.fragment.app.FragmentActivity;
import android.os.Bundle;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.google.android.gms.maps.model.PolylineOptions;
public class MainActivity extends FragmentActivity implements OnMapReadyCallback {
    private GoogleMap mMap;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
                .findFragmentById(R.id.map);
        if (mapFragment != null) {
            mapFragment.getMapAsync(this);
        }
    }
    @Override
    public void onMapReady(GoogleMap googleMap) {
        mMap = googleMap;

        // Example route: Home -> Bus Stop -> Office
        LatLng home = new LatLng(28.6139, 77.2090);     // Delhi example
```

```
            LatLng busStop = new LatLng(28.6200, 77.2100);
            LatLng office = new LatLng(28.6250, 77.2150);
            // Add markers
            mMap.addMarker(new MarkerOptions().position(home).title("Home"));
            mMap.addMarker(new MarkerOptions().position(busStop).title("Bus Stop"));
            mMap.addMarker(new MarkerOptions().position(office).title("Office"));
            // Draw route with Polyline
            mMap.addPolyline(new PolylineOptions()
                    .add(home, busStop, office)
                    .width(8)
                    .color(0xFF2196F3)); // Blue
            // Move camera to Home
            mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(home, 14));
        }
    }
```

Output :

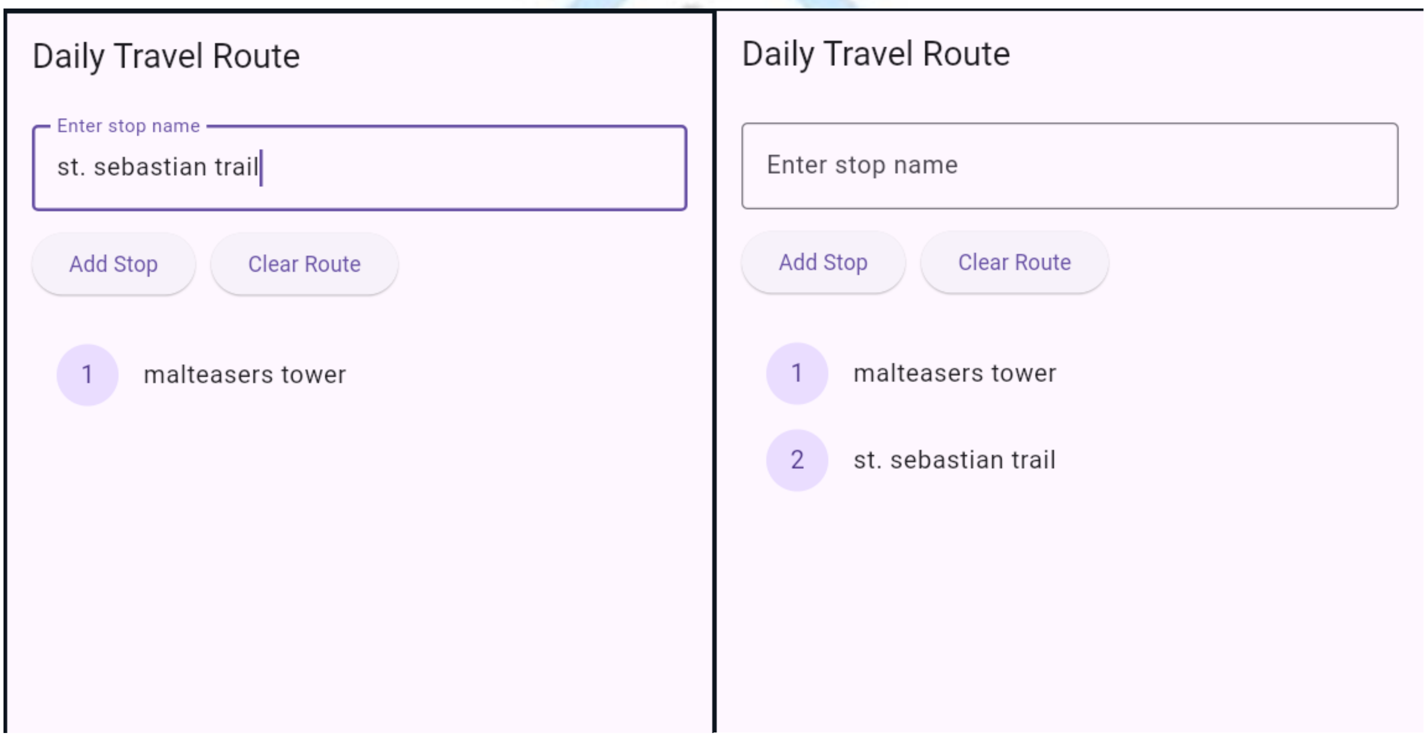

Fig 11.1 Output

Logical Application

1. **Which Android class is used to create animations on a View?**

2. **What method changes the color of a shape drawable programmatically?**

Date:

PRACTICAL 12

AIM: Write an Android application to convert into different currencies.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/etRupees"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter amount in Rupees"
        android:inputType="numberDecimal"
        android:layout_marginBottom="15dp"/>

    <Button
        android:id="@+id/btnConvert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert to Dollar" />

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result will appear here"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.currencyconverter;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    EditText etRupees;
    Button btnConvert;
    TextView tvResult;

    // Conversion rate (Example: 1 Rupee = 0.012 USD)
    final double DOLLAR_RATE = 0.012;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etRupees = findViewById(R.id.etRupees);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String rupeesStr = etRupees.getText().toString();

                if (rupeesStr.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Please enter amount in Rupees",
Toast.LENGTH_SHORT).show();
                    return;
                }

                double rupees = Double.parseDouble(rupeesStr);
                double dollars = rupees * DOLLAR_RATE;

                tvResult.setText("USD: $" + String.format("%.2f", dollars));
            }
        });
    }
}
```

Output:

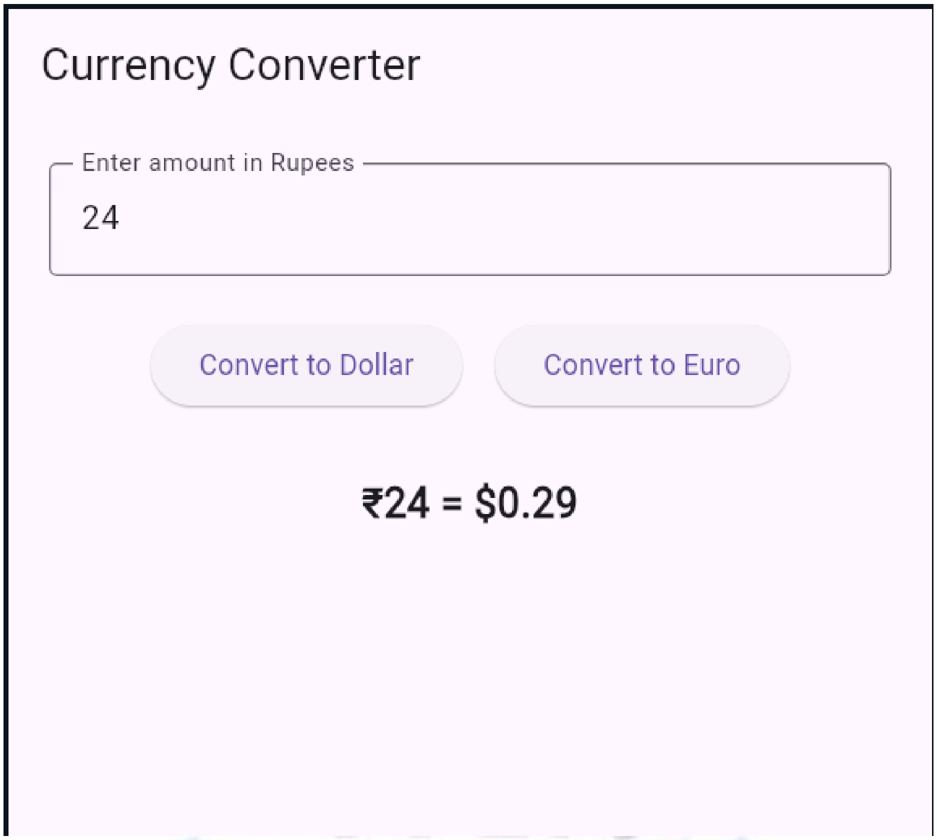

Fig 12.1 Output

Logical Application

1. **Which Google Maps API method is used to draw a path on a map?**

____
____
____

2. **What permission is required in AndroidManifest to access the device location?**

____
____
____

Date:

PRACTICAL 13

AIM: Write an application to record video and audio on topic "Intent" and play the audio and video.

Program:

AndroidManifest.xml

```
<uses-permission android:name="android.permission.CAMERA"/>
<uses-permission android:name="android.permission.RECORD_AUDIO"/>
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

<application
    ...>
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>
</application>
```

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">

    <Button
        android:id="@+id/btnRecordAudio"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Audio"/>

    <Button
        android:id="@+id/btnRecordVideo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Video"
        android:layout_marginTop="15dp"/>
```

```
    <Button
        android:id="@+id/btnPlayAudio"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Audio"
        android:layout_marginTop="15dp"/>

    <Button
        android:id="@+id/btnPlayVideo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play Video"
        android:layout_marginTop="15dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```java
package com.example.intentmediaapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    Button btnRecordAudio, btnRecordVideo, btnPlayAudio, btnPlayVideo;
    Uri audioUri, videoUri;
    final int AUDIO_REQUEST = 1;
    final int VIDEO_REQUEST = 2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnRecordAudio = findViewById(R.id.btnRecordAudio);
        btnRecordVideo = findViewById(R.id.btnRecordVideo);
        btnPlayAudio = findViewById(R.id.btnPlayAudio);
        btnPlayVideo = findViewById(R.id.btnPlayVideo);
```

```
        // Record Audio
        btnRecordAudio.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                Intent audioIntent = new Intent(MediaStore.Audio.Media.RECORD_SOUND_ACTION);
                try {
                    startActivityForResult(audioIntent, AUDIO_REQUEST);
                } catch (Exception e) {
                    Toast.makeText(MainActivity.this, "No app found to record audio",
Toast.LENGTH_SHORT).show();
                }
            }
        });

        // Record Video
        btnRecordVideo.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                Intent videoIntent = new Intent(MediaStore.ACTION_VIDEO_CAPTURE);
                try {
                    startActivityForResult(videoIntent, VIDEO_REQUEST);
                } catch (Exception e) {
                    Toast.makeText(MainActivity.this, "No app found to record video",
Toast.LENGTH_SHORT).show();
                }
            }
        });

        // Play Audio
        btnPlayAudio.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (audioUri != null) {
                    Intent playAudio = new Intent(Intent.ACTION_VIEW);
                    playAudio.setDataAndType(audioUri, "audio/*");
                    startActivity(playAudio);
                } else {
                    Toast.makeText(MainActivity.this, "No audio recorded", Toast.LENGTH_SHORT).show();
                }
            }
        });

        // Play Video
```

```
        btnPlayVideo.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (videoUri != null) {
                    Intent playVideo = new Intent(Intent.ACTION_VIEW);
                    playVideo.setDataAndType(videoUri, "video/*");
                    startActivity(playVideo);
                } else {
                    Toast.makeText(MainActivity.this, "No video recorded", Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        super.onActivityResult(requestCode, resultCode, data);
        if (resultCode == RESULT_OK) {
            if (requestCode == AUDIO_REQUEST) {
                audioUri = data.getData();
            } else if (requestCode == VIDEO_REQUEST) {
                videoUri = data.getData();
            }
        }
}
```

Output :

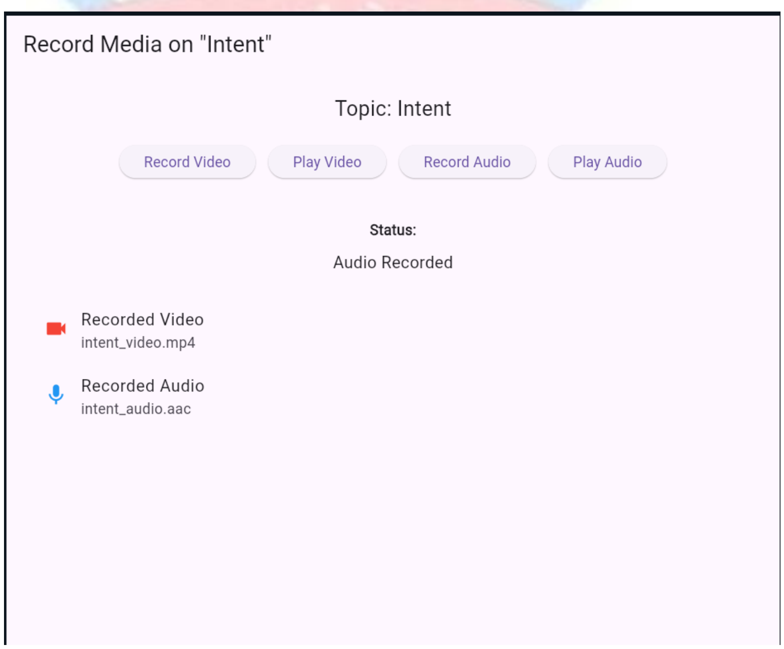

Fig 13.1 Output

Logical Application

1. **Which class in Android is used to record audio and video?**

2. **What method of MediaPlayer is used to specify the file path of media to play?**

Date:

PRACTICAL 14

AIM: Write android program to set the wallpaper of your device using bitmap class.

Program:

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
    <Button
        android:id="@+id/btnSetWallpaper"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Set Wallpaper"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.wallpaperapp;

import androidx.appcompat.app.AppCompatActivity;
import android.app.WallpaperManager;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import java.io.IOException;
public class MainActivity extends AppCompatActivity {
    Button btnSetWallpaper;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnSetWallpaper = findViewById(R.id.btnSetWallpaper);
        btnSetWallpaper.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```
                // Load bitmap from drawable
                Bitmap bitmap = BitmapFactory.decodeResource(getResources(),
R.drawable.wallpaper_image);
                // Set wallpaper using WallpaperManager
                WallpaperManager wallpaperManager =
WallpaperManager.getInstance(getApplicationContext());
                try {
                    wallpaperManager.setBitmap(bitmap);
                    Toast.makeText(MainActivity.this, "Wallpaper Set Successfully",
Toast.LENGTH_SHORT).show();
                } catch (IOException e) {
                    e.printStackTrace();
                    Toast.makeText(MainActivity.this, "Failed to Set Wallpaper",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

Output :

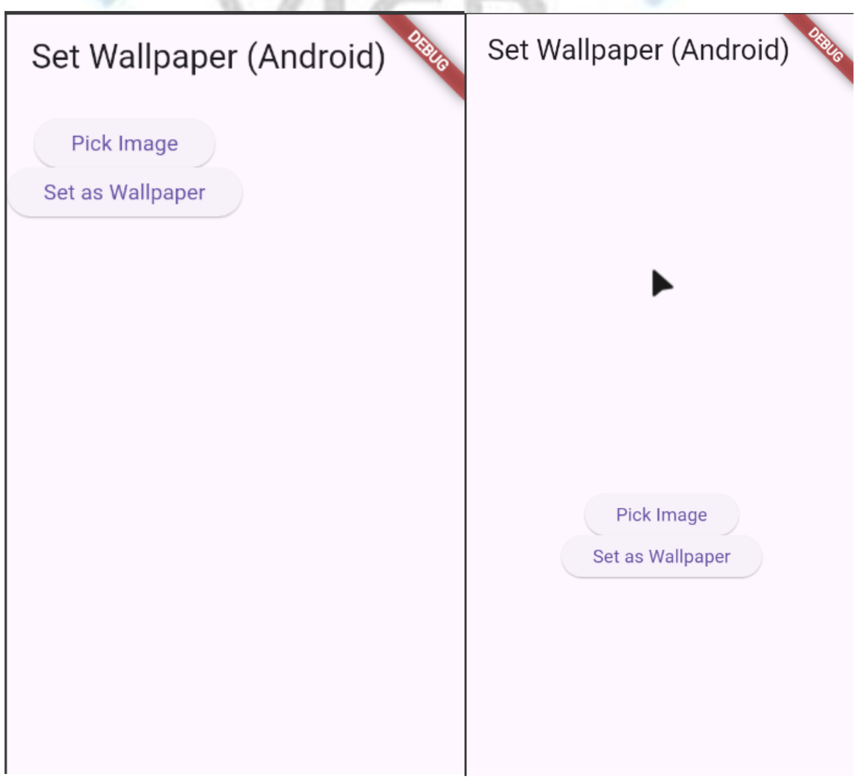

Fig 14.1 Output

Logical Application

1. **Which Android class provides the method to set the device wallpaper?**

2. **What permission must be declared in the manifest to change the wallpaper?**